



WHERE and LIKE Queries

Short Documentation: WHERE and LIKE Queries

The following script demonstrates **data filtering techniques in MySQL** using the WHERE clause, comparison operators, and the LIKE operator across the techsphere_solutions database.

1. Database Selection

```
USE techsphere_solutions;
```

What it does:

- Selects the database to be used
- Ensures all queries run on the correct schema

2. WHERE Clause Basics (Numeric & Text Filtering)

2.1 Salary greater than 90,000

```
SELECT *  
FROM job_roles  
WHERE base_salary > 90000;
```

What it does:

- Retrieves job roles where salary is above 50,000

2.2 Salary $\geq$ 90,000

```
SELECT *  
FROM job_roles  
WHERE base_salary >= 90000;
```

**What it does:**

- Includes roles with salary equal to or above 50,000

2.3 Specific department

```
SELECT *  
FROM departments  
WHERE department_name = 'Cybersecurity';
```

What it does:

- Retrieves only the Cybersecurity department record

2.4 Excluding a department

```
SELECT *  
FROM departments  
WHERE department_name != 'Cybersecurity';
```

What it does:

- Returns all departments except Cybersecurity

3. WHERE with Dates**3.1 Employees hired after a date**

```
SELECT *  
FROM employees  
WHERE hire_date > '2022-01-01';
```

What it does:

- Filters employees hired after 1st Jan 2022



3.2 Employees born after a date

```
SELECT *  
FROM employees  
WHERE date_of_birth > '1995-01-01';
```

What it does:

- Filters younger employees based on birth date

4. LIKE Operator (Pattern Matching)

4.1 Names starting with B

```
WHERE first_name LIKE 'B%'
```

What it does:

- Finds employees whose first name begins with "B"

4.2 Names starting with A and 4 characters long

```
WHERE first_name LIKE 'A____'
```

What it does

- Matches names starting with A and exactly 4 characters long

4.3 Email contains "techsphere"

```
WHERE email LIKE '%techsphere%'
```

What it does:

- Finds emails that contain the word "techsphere" anywhere



4.4 Last names starting with O

```
WHERE last_name LIKE 'O%'
```

What it does:

- Retrieves employees whose last name starts with "O"

5. Combined Conditions

5.1 AND condition

```
WHERE department_id = 1  
AND hire_date > '2021-01-01';
```

What it does:

- Returns employees who satisfy BOTH conditions:
 - Belong to department 1
 - Hired after 2021

5.2 OR condition

```
WHERE department_id = 2  
OR department_id = 3;
```

What it does:

- Returns employees in either department 2 or 3

6. Overall, Purpose of the Script

This script demonstrates how SQL is used to:

- Filter data using conditions (WHERE)
- Perform comparisons (>, >=, !=)
- Work with dates for time-based analysis
- Search text patterns using LIKE
- Combine multiple conditions using AND / OR



7. Key Learning Outcome

After running these queries, you understand how to:

- Extract specific records from large datasets
- Apply business rules in SQL queries
- Perform basic HR and analytics filtering
- Build foundational SQL query logic used in reporting and dashboards

8. Summary Insight

The WHERE clause is used for precise row filtering, while LIKE extends filtering to flexible pattern-based text searches. Combined, they form the foundation of SQL data querying and analysis.